

Kanghwi Lee

Ph.D. Candidate | Machine Learning for Real-World Audio and Biological Signals

Institute of Neuroinformatics, ETH Zurich

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Research Profile

Applied ML researcher and ETH Zurich Ph.D. candidate building scalable models and data workflows for real-world audio and biological signals. Research spans representation learning, foundation model adaptation, large-scale audio data curation, and generative modeling, grounded in computational neuroscience.

Selected Research

Large-Scale Audio Representation Workflow and Human-in-the-Loop Clustering

First-author paper, *Applied Sciences*, 2026 | [Paper](#) | [Code](#)

- Developed a semi-automated workflow using Whisper encoder embeddings, UMAP-based clustering, and interactive expert review to organize real-world vocalization data.
- Achieved $ARI = 0.73$ and $NMI = 0.81$ on a 1,454-vocalization benchmark while reducing annotation time by approximately 50% (90 to 45 minutes per bird).
- Applied the workflow to approximately 80,000 hours of recordings for downstream doctoral research.

WhisperSeg: Whisper-Based Segmentation for Human and Animal Vocal Activity

Co-authored paper, *ICASSP*, 2024 | [Paper](#)

- Adapted a pretrained Whisper Transformer for long-context, token-based VAD, generating onset, offset, and vocalization-type tokens directly from audio spectrograms across human and animal recordings.
- Achieved segment-wise $F1 = 0.9966$ on adult zebra finch recordings and outperformed frame-wise baselines across evaluated species, including noisy marmoset and human recordings.
- Contributed to dataset construction and baseline evaluation, then applied WhisperSeg segmentation outputs in subsequent large-scale vocalization analysis workflows.

Counterfactual Audio Generation and Trajectory Variance

Submitted for review to *Interspeech 2026* | [Code](#)

- Designed and implemented a PyTorch-based age-conditioned counterfactual model using VAE latent representations and mini-batch optimal-transport matching, with single-forward-pass latent displacement prediction.
- Evaluated on approximately 680K vocalizations across three zebra finches, outperforming Gaussian OT, per-age k-NN, and per-age OT baselines; Cohen's $d = 0.29\text{--}0.57$ and $AUC = 0.58\text{--}0.67$ after controlling for duration.

Computational Modeling of Vocal Template Formation

COSYNE, 2024 | [Conference Abstract](#)

- Developed a Bayesian causal-inference model of composite song-template formation from sequential tutoring, fitted to behavioral data from 19 juvenile zebra finches (26 syllables).

Publications and Conference Abstracts

Journal and Conference Papers

- Kanghwi Lee**, Maris Basha, Anja T. Zai, and Richard H. R. Hahnloser. "Benchmarking Automated and Semi-Automated Vocal Clustering Methods." *Applied Sciences*, 16, 810, 2026. [DOI](#)
- Nianlong Gu, **Kanghwi Lee**, Maris Basha, Sumit Kumar Ram, Guanghao You, and Richard H. R. Hahnloser. "Positive Transfer of the Whisper Speech Transformer to Human and Animal Voice Activity Detection." *ICASSP*, 2024. [DOI](#)

Manuscript Submitted for Review

- Kanghwi Lee**. "Trajectory Variance: An Unsupervised Measure of Developmental Vocal Plasticity." Submitted for review to *Interspeech 2026*.

Conference Abstract

- Kanghwi Lee**, Richard H. R. Hahnloser, Hazem Toutounji, and Dina Lipkind. "Serial Tutoring Reveals a Composite Song Template." *COSYNE*, 2024.

Education

ETH Zurich – Ph.D. Candidate, Institute of Neuroinformatics	/ Apr 2022 – Present; expected early 2027
University of Zurich – M.Sc., Neural Systems and Computation	/ Feb 2020 – Feb 2022; GPA 5.5/6.0
Korea University – B.S., Computer Science; B.Econ., Statistics	/ Mar 2014 – Feb 2020; GPA 4.02/4.50

Technical Skills

ML and Generative Modeling: PyTorch, Transformers, Whisper, representation learning, VAEs, flow matching, diffusion models, optimal transport

Audio and Data Workflows: audio segmentation/VAD, learned audio representations, spectrogram analysis, clustering, interactive annotation, large-scale audio processing

Programming and Computing: Python, NumPy/SciPy, scikit-learn, librosa, MATLAB, Git, Linux, HPC, parallel computing, GPU-based batch processing